

Linear Z-score: calculation sheet

GEC-CCP electron density n_e · current interp-train statistics

1 Fit on Train

$$\mu = \frac{\sum m_i n_{e,i}}{\sum m_i}$$
$$\sigma = \sqrt{\frac{\sum m_i (n_{e,i} - \mu)^2}{\sum m_i}}$$

$$\mu = 1.2915 \times 10^{15} \text{ m}^{-3}$$

$$\sigma = 1.8713 \times 10^{15} \text{ m}^{-3}$$

$m_i = 1$ in plasma, 0 outside
54 cases · 1.32 M plasma cells

2 Standardize

$$z_e = \frac{n_e - \mu}{\sigma}$$

$$n_e = 3.00 \times 10^{15}$$

$$z_e = \frac{3.00 - 1.2915}{1.8713} = 0.913$$

numbers shown in 10^{15} m^{-3}

same ordering · new center and scale

3 Return to physics

$$\hat{n}_e = \mu + \sigma \hat{z}_e$$

model prediction $\hat{z}_e$



physical $\hat{n}_e$ (m^{-3})

identity physical values · train only · plasma only · no clipping · population std (ddof = 0)

Z-score standardization does not make a skewed distribution Gaussian.